

Original article

Therapeutic effect of modulating the NLRP3-regulated transforming growth factor- β signaling pathway on interstitial cystitis/bladder pain syndromeHung-Jen Shih^{a,b,c}, Chao-Yuan Chang^{d,e,f}, Chung-Howe Lai^{a,1}, Chun-Jen Huang^{d,e,f,*}^a Department of Urology, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan^b Department of Urology, School of Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan^c Department of Urology, Changhua Christian Hospital, Changhua, Taiwan^d Integrative Research Centre for Critical Care, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan^e Department of Anesthesiology, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan^f Graduate Institute of Clinical Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan

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ABSTRACT

Interstitial cystitis/bladder pain syndrome (IC/BPS) is a disorder with complex pathogenesis and lacks effective treatment. Chronic inflammation is the main pathogenesis of Hunner-type IC/BPS. The NLR family pyrin domain-containing 3 (NLRP3) inflammasome-related transforming growth factor- β (TGF- β)/Smad signaling pathway plays a crucial role in inflammation-related tissue fibrosis. Lipopolysaccharide (LPS) and protamine sulfate (LPS/PS) were instilled into the mouse bladder twice a week for 5 consecutive weeks to establish a chronic inflammation-induced IC/BPS model (LPS/PS model). Following LPS/PS treatment, curcumin (oral, 100 mg/kg; a potent NLRP3 modulator) was administered for 2 weeks in the curcumin treatment group, and normal saline was used for the sham group. Bladder function was evaluated by performing the voiding spot assay and examining the status of urothelial denudation and fibrosis in bladder tissues. The expression of NLRP3 inflammasome, interleukin-1 β , TGF- β , Smad, vimentin, and E-cadherin in bladder tissues was evaluated through immunohistochemistry staining. Results revealed that the repeated instillation of LPS/PS leads to voiding dysfunction, bladder urothelium denudation, and detrusor muscle fibrosis through the upregulation of the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway and the increased epithelial-mesenchymal transition process in bladder tissues. The downregulation of the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway in bladder tissues through curcumin effectively mitigated bladder injury in the LPS/PS model. In conclusion, the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway plays a crucial role in bladder injury in the LPS/PS model, and modulation of this pathway, such as by using curcumin, can effectively mitigate the sequelae of chronic inflammation-induced IC/BPS.

1. Introduction

Interstitial cystitis/bladder pain syndrome (IC/BPS) is a troublesome disease presenting with irritative voiding symptoms and pelvic pain and is associated with medical comorbidities such as depression and anxiety [1,2]. A high misdiagnosis rate and lack of effective treatment severely affect patients' quality of life because the precise etiology and pathophysiology of IC/BPS are not elucidated [3–5]. Several etiological theories have been proposed on the initiation of bladder injury in patients with IC/BPS [5,6]. In Hunner-type IC/BPS patients, the main histological finding is chronic inflammation in bladder tissues, and

bladder-centric symptoms are predominant [7]. In Hunner-type IC/BPS patients, chronic inflammation induces bladder urothelial denudation and troublesome voiding symptoms such as urgency and bladder pain [8,9]. Furthermore, bladder fibrosis is an inevitable consequence, and dysfunctional bladder, such as decreased bladder compliance, occurs [8,10,11]. An effective treatment to block the inflammation process in bladder tissues is a potential innovation therapy for Hunner-type IC/BPS patients.

NLR family pyrin domain-containing 3 (NLRP3) inflammasome is a key factor related to the triggering of inflammatory processes [12,13]. After NLRP3 inflammasome activation, interleukin (IL)-1 β is activated,

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and various innate immune processes are evoked [13,14]. The NLRP3 inflammation pathway has been found to be involved in the initiation of the fibrosis process through the activation of the transforming growth factor- β (TGF- β)/Smad signaling pathway [15,16]. In the bladder outlet partial obstruction model, bladder fibrosis was induced by inflammation through the activation of the NLRP3/IL-1 β pathway [17]. Thus, the role of NLRP3/IL-1 β and TGF- β /Smad pathways in Hunner-type IC/BPS should be elucidated.

Although multiple treatment options are available for IC/BPS, IC/BPS patients have higher health care costs and comorbidities than non-IC/BPS patients due to lack effective and lasting treatment options [18]. An effective therapy with a high safety profile and easy-to-use regimen is urgently needed to increase medication adherence and alleviate symptoms in IC/BPS patients. Curcumin is a polyphenol derived from turmeric (*Curcuma longa*) rhizome, which is commonly used as a spice and herb in Asia [19]. Curcumin is used as a potential therapeutic agent for several chronic diseases, especially inflammatory diseases, because of its anti-inflammatory, antioxidant, and immunomodulatory properties [19]. The NLRP3/IL-1 β pathway could be effectively inhibited by curcumin, and curcumin has great potential to treat NLRP3-induced diseases [20,21]. In animal studies, curcumin has been effectively used to treat myocardial fibrosis, liver cirrhosis, and idiopathic pulmonary fibrosis [21–23]. However, the effect of curcumin on inflammation-induced bladder injury has not been studied. We hypothesized that the downregulation of NLRP3 expression in bladder tissues by curcumin can attenuate bladder injury in Hunner-type IC/BPS. Thus, we evaluated the therapeutic effect of curcumin and its possible therapeutic mechanisms in inflammation-induced bladder injury disease by using a bladder-centric mouse model.

2. Materials and methods

2.1. Animals

In this study, we used young adult female BALB/c mice (22–26 g in weight; BioLASCO, Taipei, Taiwan). This animal study was approved by the Institutional Animal Care and Use Committee of Taipei Medical University (IAC-2019-0248). The care and handling of mice were in accordance with the guidelines of the National Institutes of Health, the United States.

2.2. Chronic inflammation-induced IC/BPS mouse model (lipopolysaccharide/protamine sulfate model) and study design

Eight-week-old female BALB/c mice were anesthetized with zoletil/xylazine (40/10 mg/kg, i.p.), and urethral catheterization was performed using a 24-gauge angiocatheter to empty the bladder and instill drugs. Subsequently, mice were treated with protamine sulfate (PS) and lipopolysaccharide (LPS). First, PS (1.5 mg/mice, Sigma-Aldrich, St. Louis, MO, USA) was instilled into the bladder and kept in the bladder for 30 min. After draining out PS, the bladder was washed with normal saline and then emptied. Then, LPS (150 μ g/mouse, Sigma-Aldrich, St. Louis, MO, USA) was instilled into the bladder and kept in the bladder for 30 min. To increase the urothelial permeability and enhance the inflammatory injury of the bladder induced by LPS, PS was used before LPS instillation. PS- and LPS-treated mice received PS and LPS bladder instillation twice weekly (day 1 and day 4 per week) for 5 consecutive weeks. In the sham group, an equal volume of normal saline (100 μ L) as the volume of PS combined with LPS was used. In the treatment group, 100 mg/kg curcumin (ChemCruz, Huissen, the Netherlands; oral gavage) was administered for 5 consecutive days, and no treatment was administered for the next 2 days. The regimen was followed for 2 consecutive weeks.

To confirm the functional and histological results of the LPS/PS model in mice, PS with LPS treatment (LPS group, $n = 3$) and sham procedure (sham group, $n = 3$) were used. In the following formal study,

curcumin alone (Cur group) and LPS plus curcumin (LC group) were added. In total, four groups ($n = 3$ in sham and Cur groups and $n = 6$ in LPS and LC groups) were used to evaluate the role of NLRP3-induced fibrosis in the LPS/PS model (Fig. 1).

2.3. Micturition assessment

After complete treatment, the void spot assay was performed to evaluate the micturition pattern of mice as described previously [24]. Mice (one mouse per cage) were placed in a cage containing standard rodent chow but no water and lined with 0.35-mm filter paper (26.5 $\times$ 21 cm; No. GFP001; GenePure, Taichung, Taiwan) for 4 h in a dark room. Urine spots on filter papers were imaged using a UVP UVsolo touch darkroom ultraviolet imaging cabinet (Analytik, Jena, Germany). Image capture settings were adjusted using the UVP VisionWorks touch image acquisition software. Images were captured by performing the ninhydrin test (1% ninhydrin in 75% EtOH solution) and through 365-nm UV illumination. Exposure settings were optimized to maximize the signal over noise. The numbers and areas of urine spots were recorded.

The bladder capacity was measured as the amount of the maximal instilled normal saline volume. A 24-gauge angiocatheter was inserted into the bladder, and the bladder was infused with normal saline at a rate of 0.2 mL/min. The infusion was stopped when urine leakage occurred, and the total infusion volume was recorded as the bladder capacity.

2.4. Evaluation of histologic changes

The bladder urothelium integrity was evaluated through the immunohistochemistry (IHC) staining of cytokeratin. The 5- μ m paraffin sections of bladder tissues were incubated with the primary antibody of cytokeratin (dilution; ab9377, Abcam, Cambridge, UK). The staining procedures were conducted using a Ventana Benchmark XT autostainer (Ventana Medical Systems, Tucson, AZ, USA) according to manufacturer's instructions. All sections were observed under the TissueGnostics Axio Observer Z1 microscope (TissueGnostics GmbH, Vienna, Austria).

The bladder tissues were stained with Masson's trichrome stain by using the Trichrome Stain Kit (Sigma-Aldrich, St. Louis, MO, USA) to assess the fibrosis status as described previously [25]. Random six fields from each sample in three different layers (urothelium, lamina propria, and muscle) were selected to quantify the density of Masson's trichrome staining, and the density was examined using the ImageJ software, version 1.52e. The average value of the staining density of six fields in each section was recorded.

2.5. Evaluation expression of NLRP3 inflammasome and IL-1 β in bladder tissues

The expression of the inflammasome components NLRP3 and apoptosis-associated speck-like protein containing a caspase recruitment domain (ASC) and IL-1 β in bladder tissues was observed through IHC staining to evaluate the inflammatory status of bladder tissues. The 5- μ m paraffin sections of bladder tissues were incubated with the primary antibodies of NLRP3 (dilution 1:200; ab214185, Abcam, Cambridge, UK), ASC (dilution 1:200; NBP1-78977, Novus Biologicals, Littleton, CO, USA), and IL-1 β (dilution 1:50; AF-401-NA, R&D Systems, Minneapolis, MN, USA) as described previously [26]. Random six fields from each sample in three different layers (bladder urothelium, lamina propria, and muscle) were selected to quantify the density by using ImageJ software, version 1.52e. The average value of the staining density of six fields in each section was recorded.

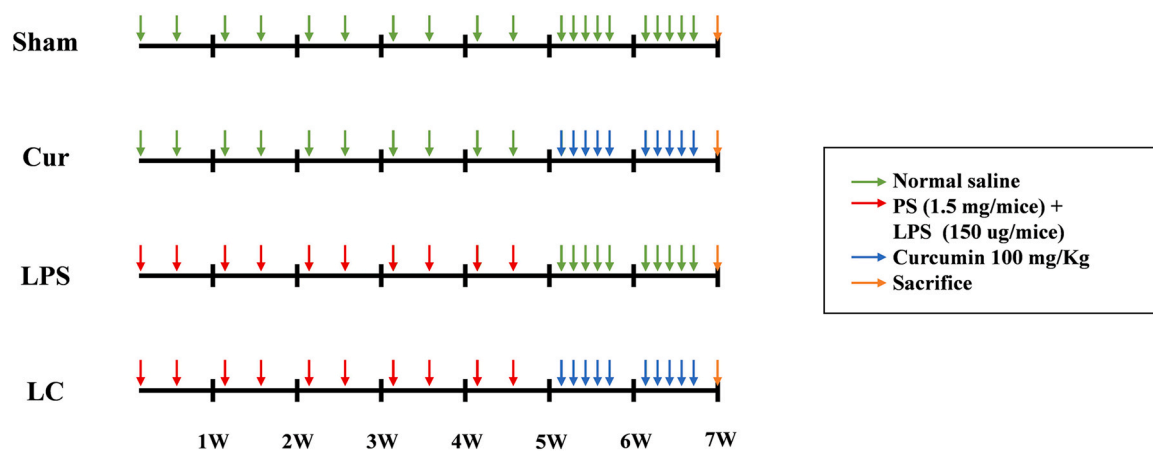


Fig. 1. Schematic diagram of the experimental design.

2.6. Evaluation expression of TGF- β /Smad signaling and epithelial-mesenchymal transition in bladder tissues

The TGF- β /Smad signaling and epithelial-mesenchymal transition (EMT) in bladder tissues were evaluated through IHC staining to determine the role of TGF- β /Smad signaling in inflammation-induced bladder fibrosis. The 5- μ m paraffin sections of bladder tissues were incubated with the primary antibodies of TGF- β 1 (dilution 1:100; ab92486, Abcam, Cambridge, UK), p-Smad2 (dilution 1:100; 44-244G, Invitrogen, Carlsbad, CA, USA), p-Smad3 (dilution 1:100; 44-246G, Invitrogen, Waltham, MA, USA), vimentin (dilution 1:500; AX10005, Abcepta, San Diego, CA, USA), and E-cadherin (dilution 1:250; ab76055, Abcam, Cambridge, UK) as described previously [27,28]. The densities of each marker were evaluated through ImageJ in the urothelium, lamina propria, and muscle. The average value of the staining density of six fields in each section was recorded.

2.7. Immunoblotting assay

For detection of NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway and EMT relative proteins in bladder tissues, immunoblotting assay was performed. Snap frozen bladder tissues were thawed, homogenized, microwaved, centrifuged and then the supernatants were collected for analyses. Protein concentration of each sample was measured (bicinchoninic acid protein assay kit, Pierce Chemical Co., Rockford, IL, USA). Equal amount proteins were separated by electrophoresis and then transferred to nitrocellulose membranes (Bio-Rad Laboratories, Hercules, CA, USA). Then, the membranes were incubated overnight with primary antibody of NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway-related proteins (including NLRP3, IL-1 β , TGF- β 1, p-Smad2, p-Smad3), EMT-related proteins (vimentin, E-cadherin), or actin (as the internal standard; Sigma-Aldrich, St. Louis, MO, USA). Bound antibody was detected by chemiluminescence (ECL plus kit; Amersham BioSciences, Buckinghamshire, UK). Experiments were repeated three times (Supplementary Fig. 1). The images were acquired by multispectral imaging system (UVP BioSpectrum 815 Imaging System) and analyzed by Vision Works LS software (UVP, California, USA).

2.8. Statistical analysis

Between-group differences were analyzed using the unpaired *t*-test, one-way analysis of variance, and post hoc pairwise comparisons with Tukey's test. Statistical significance was indicated if $p < 0.05$. The *prism* package in GraphPad (GraphPad Software, Inc., La Jolla, CA, USA) was used for statistical analyses.

3. Results

3.1. Chronic inflammation-induced bladder injury and voiding dysfunction in LPS/PS mouse model

The spontaneous void spot assay was used to quantify the rodent voiding function. Our results revealed a significant increase in the number of urine spots in the LPS group compared to the sham group ($p = 0.0001$; Fig. 2A). The urine spot area of each micturition ($p = 0.0005$) was significantly decreased in the LPS group compared to the sham group (Fig. 2A). Furthermore, the bladder capacity was significantly decreased in the LPS group compared to the sham group ($p = 0.0065$; Fig. 2B). The results of the voiding function assay demonstrated that inducing chronic inflammation using LPS/PS significantly exacerbated storage symptoms because of a decreased bladder capacity.

The bladder urothelial defect plays a crucial role in irritative voiding symptoms. Cytokeratin IHC was used to evaluate the integrity of the bladder urothelium. After LPS/PS treatment for 5 consecutive weeks, the defect of the bladder urothelium in a broad area was observed in the LPS group. A decreased bladder capacity with an increased urinary frequency was related to the results of fibrosis-induced poor bladder compliance [29]. We further evaluated the fibrosis status of the bladder wall with Masson's trichrome staining. Strong staining of submucosal and muscle layers was observed in the LPS group, and light staining was noted in the sham group. The intensities of Masson's trichrome staining in submucosal and muscle layers were significantly higher in the LPS group compared to the sham group ($p < 0.0001$ in submucosal and muscle layers; Fig. 2C). In summary, the typical histological findings of chronic inflammation-induced IC/BPS, such as bladder urothelial denudation and bladder wall fibrosis, were observed in the LPS/PS model.

3.2. NLRP3-related TGF- β /Smad signaling pathway involved in the pathogenesis of bladder epithelial-mesenchymal transition in the LPS/PS mouse model

NLRP3 inflammasome and IL-1 β activation are important priming steps for tissue fibrosis [30]. An increased expression of NLRP3 and ASC was observed in the bladder urothelium but not in submucosal or muscle layers in the LPS group compared with the sham group. Furthermore, the intensities of NLRP3 and ASC in the bladder urothelium were significantly increased in the LPS group compared to the sham group ($p = 0.0003$ for NLRP3 and $p = 0.0005$ for ASC; Fig. 3). The expression of IL-1 β was similar to that of NLRP3 and ASC in the bladder lamina propria of the LPS group. The intensity of IL-1 β in the bladder lamina propria was significantly increased in the LPS group compared to the

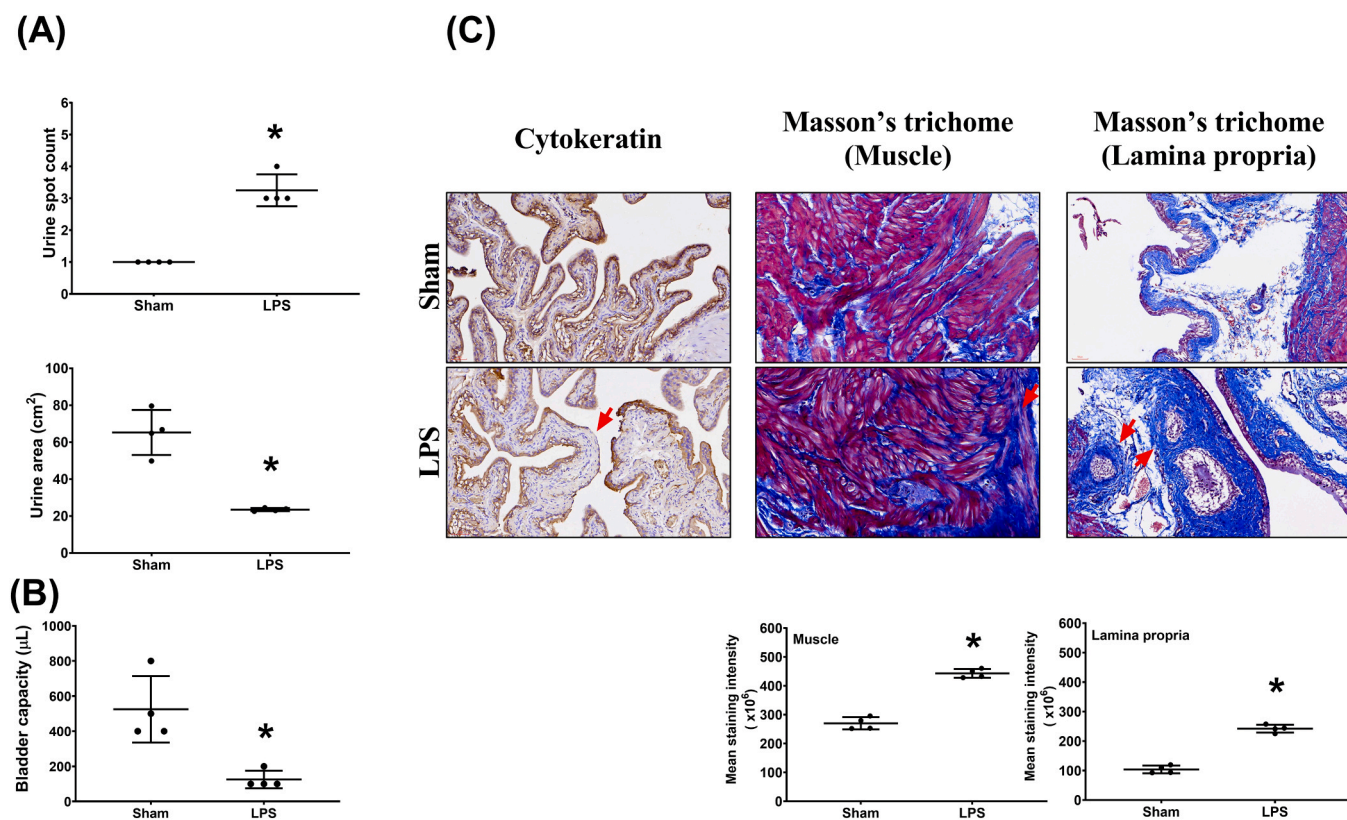


Fig. 2. Representative results of voiding function and histologic changes in the LPS/PS-induced IC/BPS mice model. (A) Void spot assay. (B) Bladder capacity. (C) Immunohistochemistry of cytokeratin and Masson's trichrome in bladder tissues. Quantification of Masson's trichrome expression in bladder tissues represented by the mean staining intensity. Arrowhead in cytokeratin staining indicated the urothelium denudation area. Arrowhead in Masson's trichrome staining pointing the fibrosis area in muscle and lamina propria. * $p < 0.05$ versus the sham group. 20× magnification.

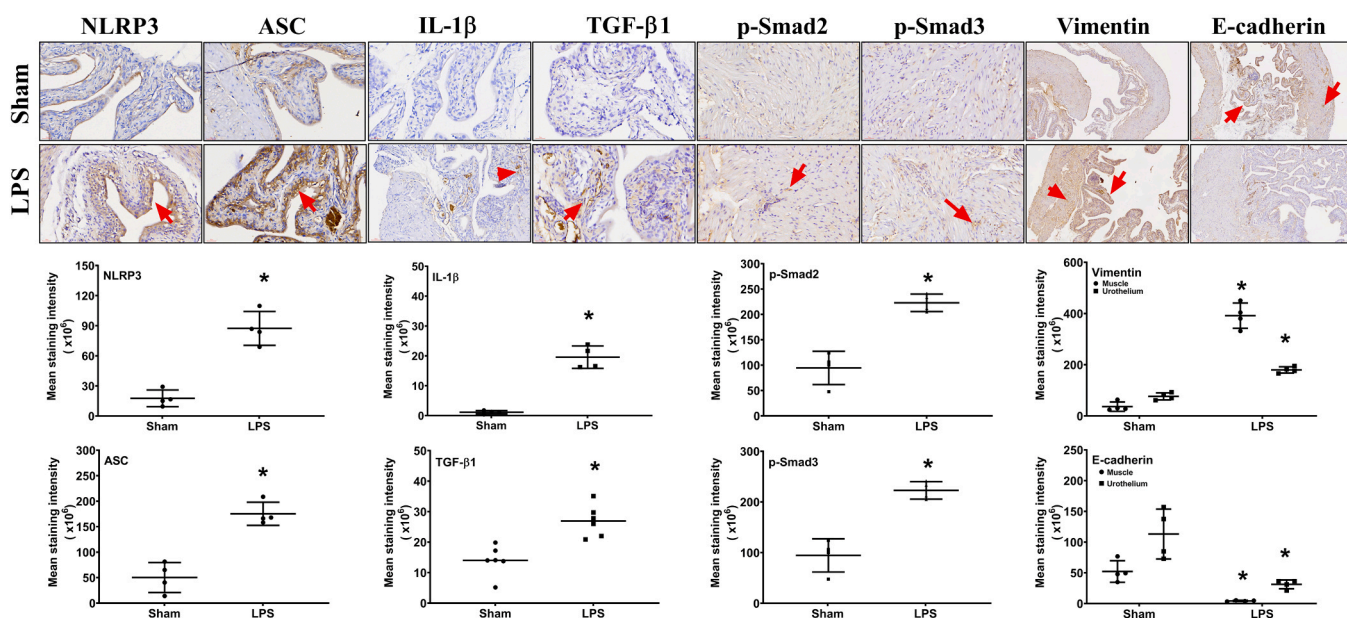


Fig. 3. Representative results of the immunohistochemistry of NLRP3-related TGF-β/Smad signaling pathway and epithelial-mesenchymal transition in the LPS/PS-induced IC/BPS mice model. Quantification of the intensity of immunohistochemistry staining in bladder tissues represented by the mean staining intensity. Arrowhead indicates the area of increased staining intensity. * $p < 0.05$ versus the sham group. 20× magnification for NLRP3, ASC, IL-1β, TGF-β, p-Smad2, and p-Smad3; 10× magnification for vimentin and E-cadherin.

sham group ($p < 0.0001$; Fig. 3). The results of the IHC study of NLRP3 inflammasome and IL-1β revealed that the inflammatory cascade was initiated in the bladder urothelium through the upregulation of the

expression of NLRP3 inflammasome and IL-1β in the LPS/PS model.

To evaluate the role of TGF-β/Smad signaling in bladder tissues, we examined TGF-β1 and p-Smad 2/3 expression through IHC staining. The

increased expression of TGF- β 1 after LPS/PS treatment was observed in the lamina propria; TGF- β 1 expression in the lamina propria was lower in the sham group (Fig. 3). The intensity of TGF- β 1 was significantly higher in the LPS group compared to the sham group ($p = 0.0165$; Fig. 3). The expression of p-Smad2 and p-Smad3 was higher in the muscle layer of the LPS group compared to the sham group (Fig. 3). The intensities of p-Smad2 and p-Smad3 were significantly higher in the LPS group compared to the sham group ($p = 0.0002$ for p-Smad2 and $p = 0.0004$ for p-Smad3; Fig. 3). In summary, these data demonstrated that chronic inflammation increased TGF- β 1 expression in the submucosa and p-Smad 2/3 in the detrusor muscle following the activation of NLRP3 inflammasome in the urothelium of the LPS/PS model.

EMT initiation is an important step of tissue fibrosis following TGF- β activation [31]. The EMT status was evaluated through the IHC staining of vimentin and E-cadherin in bladder tissues. Our results demonstrated that LPS induced vimentin expression and inhibited E-cadherin expression in urothelial and muscle layers (Fig. 3). The intensity of vimentin in the urothelium and detrusor muscle significantly increased after LPS treatment compared with sham treatment ($p < 0.0001$ for urothelium and muscle layers; Fig. 3). The intensity of E-cadherin in the urothelium and muscle layers was opposite to that of vimentin (Fig. 3). The histological findings of our study indicated that EMT plays a crucial role in urothelial denudation and detrusor muscle fibrosis in the LPS/PS model.

3.3. Curcumin mitigated bladder injury through the modulation of the NLRP3-related TGF- β /Smad signaling pathway in the LPS/PS mouse model

To evaluate the therapeutic potential of the modulated NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway in the LPS/PS model, curcumin was used because of its potent antifibrosis effects exerted through the repression of NLRP3 and TGF- β /Smad pathways [20,32]. After 5 consecutive weeks of LPS/PS bladder instillation, followed by 2 consecutive weeks of oral curcumin treatment, the number of urine spots significantly decreased in the LC group compared to the LPS group ($p = 0.036$; Fig. 4A). The urine spot areas of each micturition ($p = 0.0422$) and bladder capacity ($p < 0.0001$) were significantly increased in the LC group compared to the LPS group (Fig. 4A,B). The integrity of the bladder urothelium was preserved in the LC group (Fig. 4C). The intensities of Masson's trichrome staining were significantly decreased in the LC group compared to the LPS group ($p < 0.0001$ in submucosal and muscle layers; Fig. 4C). These results demonstrated that curcumin effectively mitigated bladder injury and micturition problems in the LPS/PS model.

We performed an IHC study to evaluate the role of the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway after curcumin treatment in the LPS/PS model. Our results demonstrated that the

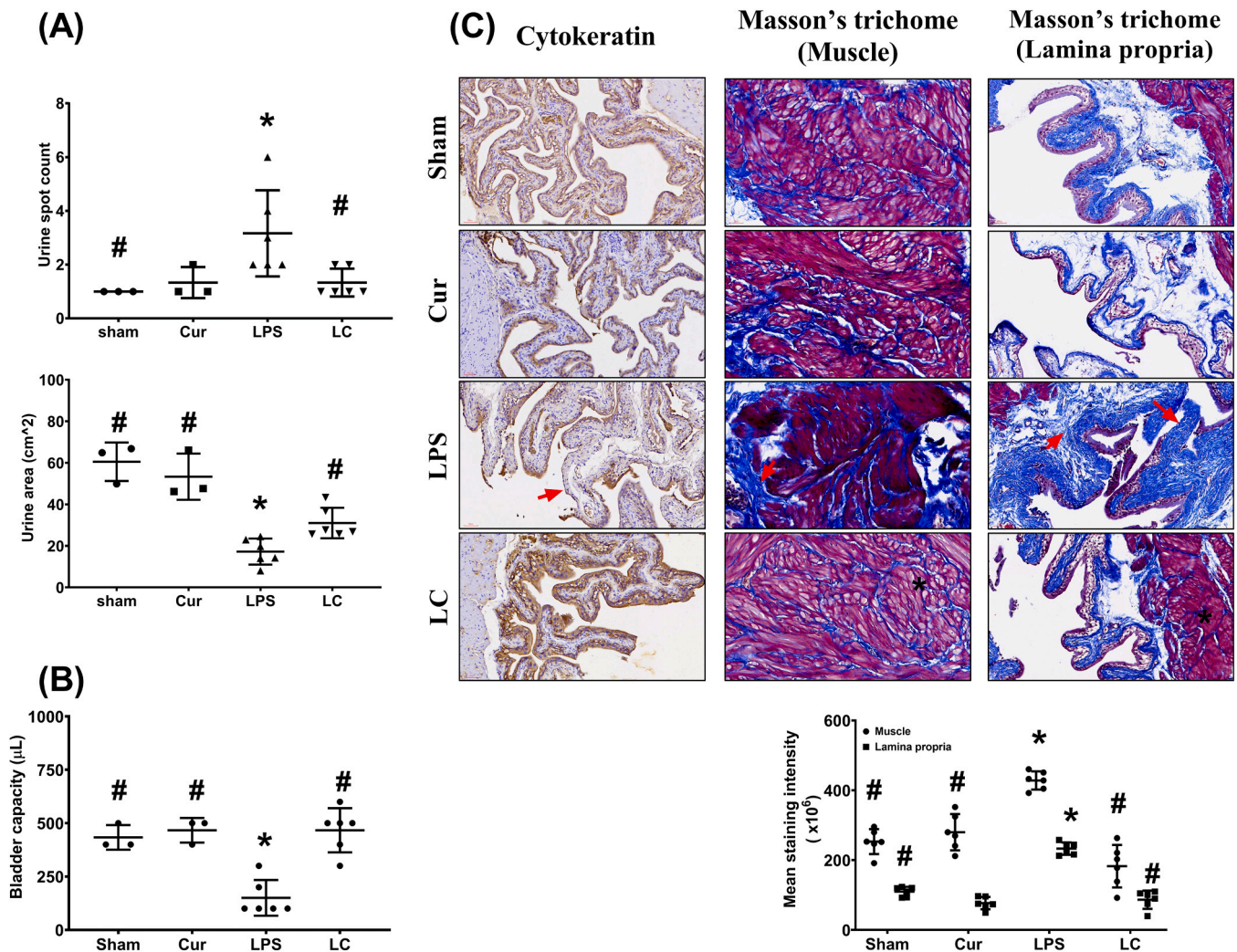


Fig. 4. Therapeutic effects of curcumin in the LPS/PS-induced IC/BPS mice model. (A) Void spot assay. (B) Bladder capacity. (C) Immunohistochemistry of cytokeratin and Masson's trichrome expression in bladder tissues. Quantification of Masson's trichrome expression represented by the mean staining intensity. Arrowhead in cytokeratin staining indicates the urothelium denudation area. Arrowhead in Masson's trichrome staining pointing the fibrosis area in muscle and lamina propria. * $p < 0.05$ versus the sham group. # $p < 0.05$ versus the LPS group. 20 $\times$ magnification.

expression of NLRP3 and ASC in the bladder urothelium decreased after curcumin treatment compared with LPS/PS treatment. The intensities of NLRP3 and ASC in the LC group were significantly decreased compared with those in the LPS group ($p = 0.0197$ for NLRP3 and $p < 0.0001$ for ASC; Fig. 5). A decreased expression of IL-1 β was observed in the bladder lamina propria after curcumin treatment compared with the LPS group, and the intensity of IL-1 β was significantly decreased in the LC group compared to the LPS group ($p < 0.0001$; Fig. 5). The expression of TGF- β 1 after LPS treatment in the lamina propria was downregulated in the LC group (Fig. 5). The TGF- β 1 intensity of the LC group was significantly lower than that of the LPS group ($p = 0.0079$; Fig. 5). The expression of p-Smad2 and p-Smad3 was lower in the detrusor muscle in the LC group (Fig. 5). The intensities of p-Smad2 ($p = 0.0005$) and p-Smad3 ($p = 0.0002$) of the LC group were significantly lower than those of the LPS group (Fig. 5). The NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway-related proteins expression in bladder tissues of the LC group were lower than LPS group (Fig. 6). Curcumin treatment reversed the expression of vimentin and E-cadherin in the urothelium and detrusor muscle according to the results of IHC staining (Fig. 5) and immunoblotting (Fig. 6). The intensity of vimentin significantly decreased in the LC group compared to the LPS group ($p < 0.0001$; Fig. 5). The intensity of E-cadherin significantly increased in the LC group compared to the LPS group ($p < 0.0001$; Fig. 5). In summary, these results demonstrated that curcumin effectively mitigated the sequelae of LPS/PS-induced bladder injury through the modulation of the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway.

4. Discussion

The etiology of IC/BPS remains complex and uncertain, and multiple animal models have been developed to evaluate the disease pathogenesis and develop a novel treatment [4,33]. However, none of the animal models could duplicate the disease course in humans [33]. Bladder chronic inflammation has been identified as the main player in the pathogenesis of Hunner-type IC/BPS, and bladder-centric models play a crucial role in evaluating possible pathogenesis and testing the therapeutic effect of medications [8,33]. A simple and reproducible bladder-centric animal model is urgently needed for researchers to study IC/BPS. Studies have reported that chronic inflammation induced by repeated instillation of LPS and PS into the rat bladder is a useful animal model for IC [34,35]. However, a chronic inflammation-induced mouse model mimicking the disease process in humans is lacking. In this bladder-centric chronic inflammation-induced bladder injury mouse model, repeated stimulation with LPS and PS produced typical histological changes, such as bladder urothelial denudation and fibrosis formation in the submucosa and detrusor muscle, and clinical presentations, such as an increase in the voiding frequency with a small-voided volume in human Hunner-type IC/BPS. This mouse model is a useful modality when experimental medication is expensive or when performing an experiment with specific-knockout mice.

NLRP3 inflammasome triggers the activation of the inflammatory cascade, and aberrant inflammasome activation can cause tissue injury [13]. In the acute stage of bladder inflammation in the cyclophosphamide-induced IC/BPS model, the activation of the inflammasome-related pathway is an important mediator of bladder inflammation. However, in a clinical scenario, the therapeutic potential of pharmacological targets in the initial clinical course of IC/BPS

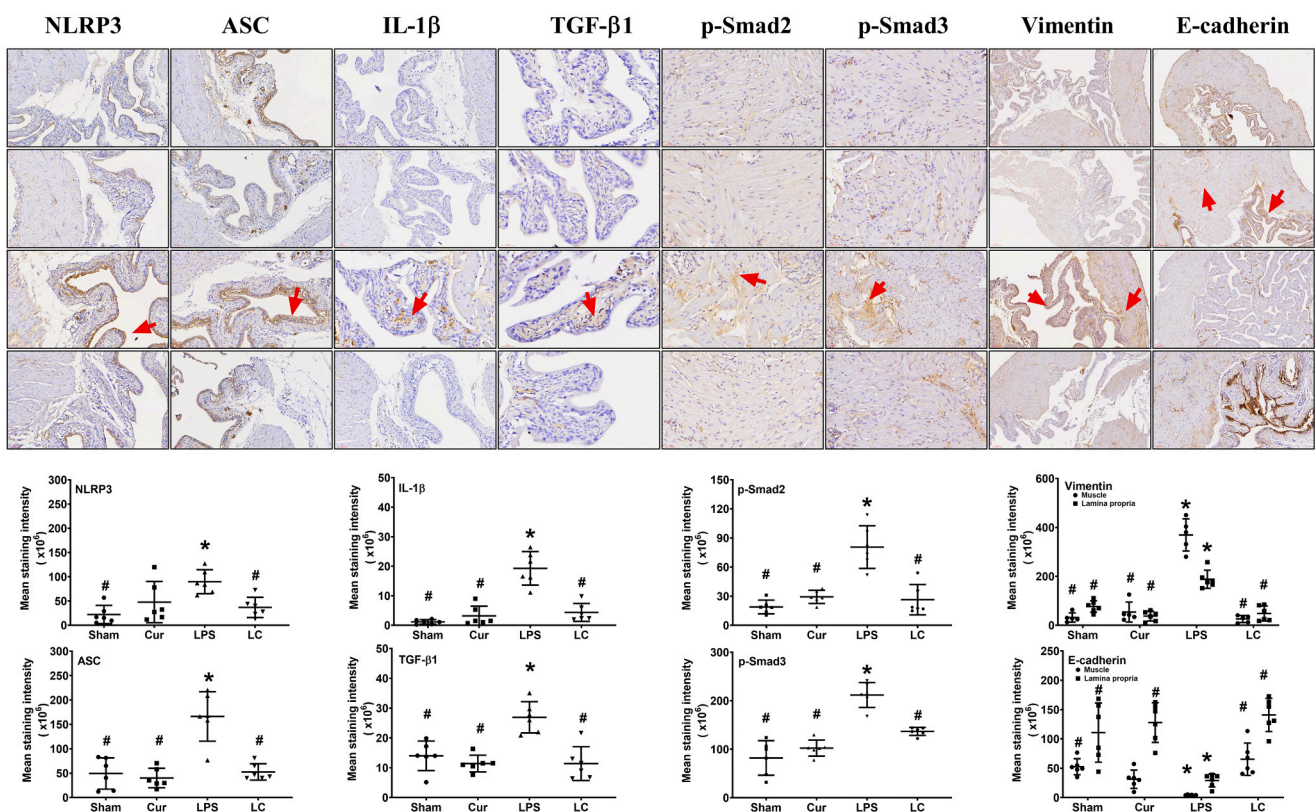


Fig. 5. Curcumin mitigates bladder epithelial-mesenchymal transition in the LPS/PS-induced IC/BPS mice model through the modulation of the NLRP3-related TGF- β pathway. Representative results of the immunohistochemistry of NLRP3-related TGF- β /Smad signaling pathway and epithelial-mesenchymal transition in bladder tissues. Quantification of immunohistochemistry staining intensity in bladder tissues represented by the mean staining intensity. Arrowhead indicates the area of increased staining intensity. * $p < 0.05$ versus the sham group. # $p < 0.05$ versus the LPS group. 20 $\times$ magnification for NLRP3, ASC, IL-1 β , TGF- β , p-Smad2, and p-Smad3; 10 $\times$ magnification for vimentin and E-cadherin.

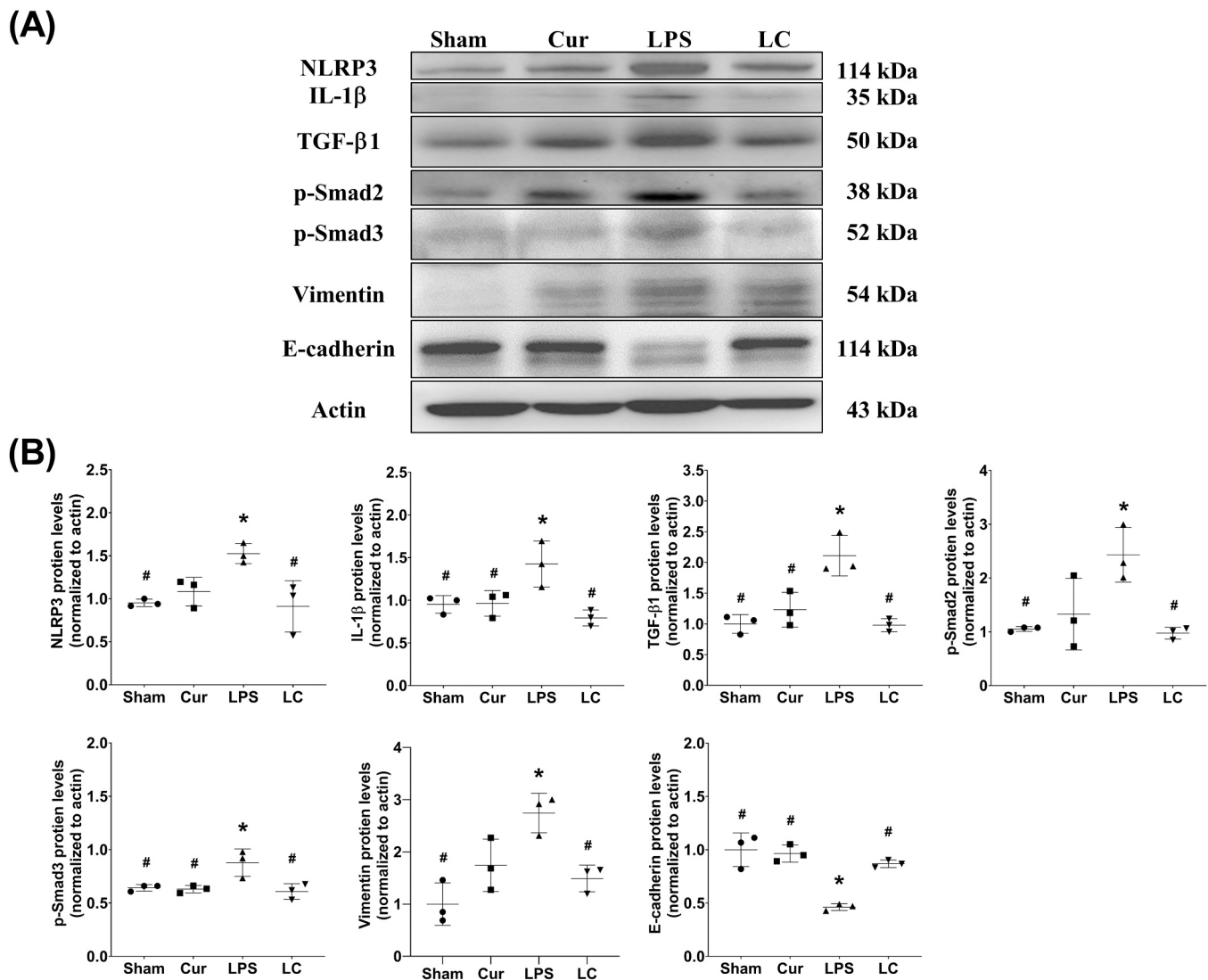


Fig. 6. Representative results of the immunoblotting of NLRP3-related TGF- β /Smad signaling pathway and epithelial-mesenchymal transition related proteins in the LPS/PS-induced IC/BPS mice model. (A) Western blot assay for the level of NLRP3, IL-1 β , TGF- β , p-Smad2, and p-Smad3, vimentin, E-cadherin and actin in bladder tissues. (B) Quantification of band intensity represented as ratio to targeting protein over actin. All results are shown as mean $\pm$ S.D. * $p < 0.05$ versus the sham group. # $p < 0.05$ versus the LPS group.

patients is limited due to delayed diagnosis. The role of inflammasome dysregulation in the chronic stage of IC/BPS remains unclear. In line with the results of the bladder outlet obstruction animal model, NLRP3 triggered downstream fibrosis-related signals through the IL-1 receptor on the bladder urothelium [17]. This study demonstrated that the expression of NLRP3 in the bladder urothelium and that of IL-1 β in the lamina propria remained upregulated after 5 weeks of LPS and PS stimulation. Therefore, the upregulation of the inflammasome-related pathway is a crucial pathogenesis mechanism of disease progression in the chronic inflammatory stage. Furthermore, a theory was proposed that NLRP3 inflammasome activation induces EMT and tissue fibrosis [15]. The results of this study revealed that inflammasome activation mediated the upregulation of TGF- β /Smad signaling and the induction of EMT. EMT occurred in the urothelium and muscle layer, and the result is compatible with the presentations of urothelial denudation and muscle fibrosis in the typical histological findings of Hunner-type IC/BPS. In summary, the pathogenesis of chronic inflammation-induced bladder injury is related to NLRP3 inflammasome and IL-1 β upregulation in the urothelium and downstream TGF- β /Smad signaling activation.

The therapeutic effect of curcumin on chronic inflammation-induced

IC/BPS mice was supported by the attenuation of irritative voiding symptoms and the histological manifestations of tissues injury. The possible therapeutic mechanisms involved the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad signaling pathway. According to the results of our study, oral curcumin administration for 2 consecutive weeks showed a favorable therapeutic potential for Hunner-type IC/BPS patients because of the easy-to-use regimen and the safety profile of curcumin [36]. In line with the results of this study, the treatment efficacy of orally administered curcumin for inflammation-related diseases, such as diabetes mellitus and cancer, has been reported owing to its potent anti-inflammatory, antioxidative, and immunomodulatory effects despite its low bioavailability on oral administration [37–40]. We postulated that the potent anti-inflammatory effect of curcumin metabolites, such as curcumin sulphate, which has been found in urine may play a crucial role in the chronic inflammation-induced IC/BPS model [41,42]. Additional studies are needed to determine the expression of curcumin metabolites in the urine and bladder tissues of the inflammation-induced IC/BPS mice. Furthermore, according to the results of this study, targeting the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad signaling pathway has a great therapeutic potential for patients with chronic inflammation-induced IC/BPS. Novel and precise

treatment other than curcumin can be developed in the future.

Our study provided the following crucial results: first, consecutive instillation of LPS and PS in the bladder for 5 weeks established a chronic inflammation-induced IC/BPS animal model; second, NLRP3 inflammasome activation initiated bladder injury through the TGF- β /Smad pathway, which plays a crucial role in the pathogenesis of chronic inflammation-induced IC/BPS; third, modulation of the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway by curcumin can be an effective therapy for chronic inflammation-induced IC/BPS. However, this study has some limitations. First, the effect on pain management of curcumin could not be assessed because pain score was not recorded in this study. Second, the effects of curcumin on the urethra or other visceral organs were not evaluated in this study. Third, the results of this study are focused on chronic inflammation-induced IC/BPS (Hunner-type IC/BPS) by using bladder-centric model. Other animal models, such as psychological stress model should be used to evaluate other etiologies related to IC/BPS. Fourth, the translation value of experimental results into humans is still limited because of the complexities of pathogenesis in IC/BPS patients. However, the animal model is still important to evaluate the pathogenesis of this disease and to search for novel treatment targets for treating this disorder.

5. Conclusion

The pathogenesis of chronic inflammation-induced IC/BPS is associated with the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway. Downregulation of the expression of the NLRP3 inflammasome/IL-1 β -related TGF- β /Smad pathway through curcumin mitigates chronic inflammation-induced bladder injury.

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Author contributions

HJS, CYC, CHL, and CJH contributed substantially to the study conception as well as to data acquisition, analysis, and interpretation, and the drafting of the paper; all authors have approved the latest version of the submitted manuscript.

HJS, CYC, CHL and CJH have agreed to be accountable for all aspects of the study and to ensure that doubts related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Declaration of conflicting interests

None declared.

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None.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.biopha.2021.111522](https://doi.org/10.1016/j.biopha.2021.111522).

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